

THE METHOD OF LAGRANGE MULTIPLIERS

Lecture 28: Presented by Jacomine Grobler



Stellenbosch Unit for Operations Research in Engineering
Department of Industrial Engineering

In this section, we consider the special case in which the objective is to

$$\text{maximise } z = f(\mathbf{x})$$

subject to the set of equality constraints

$$\begin{aligned} h_j(\mathbf{x}) &= b_j, & j = 1, \dots, q \\ \mathbf{x} &\in \mathbb{R}^n \end{aligned}$$

for some natural number q , where the functions $h_1, \dots, h_q$ are all twice differentiable with respect to all pairs of their argument components and $b_1, \dots, b_q$ are real constants.

The method of Lagrange multipliers

Lagrange suggested dualising the problem by forming a so-called *Lagrangian function*

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{j=1}^q \lambda_j [b_j - h_j(\mathbf{x})]$$

for some vector $\boldsymbol{\lambda} = [\lambda_1 \ \cdots \ \lambda_q]^T$ of coefficients, called *Lagrange multipliers*, and then considering the unconstrained non-linear optimisation problem of

$$\text{maximising } L(\mathbf{x}, \boldsymbol{\lambda}), \quad \mathbf{x} \in \mathbb{R}^n, \quad \boldsymbol{\lambda} \in \mathbb{R}^q.$$

The method of Lagrange multipliers

Note that if $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ is a local optimum of $L(\mathbf{x}, \boldsymbol{\lambda})$, then it is a stationary point of $L(\mathbf{x}, \boldsymbol{\lambda})$ (Theorem 2.4), which implies that

$$\left. \frac{\partial L}{\partial x_1} \right|_{(\mathbf{x}^*, \boldsymbol{\lambda}^*)}, \dots, \left. \frac{\partial L}{\partial x_n} \right|_{(\mathbf{x}^*, \boldsymbol{\lambda}^*)} = 0 \quad \text{and} \quad \left. \frac{\partial L}{\partial \lambda_1} \right|_{(\mathbf{x}^*, \boldsymbol{\lambda}^*)}, \dots, \left. \frac{\partial L}{\partial \lambda_q} \right|_{(\mathbf{x}^*, \boldsymbol{\lambda}^*)} = 0.$$

In particular,

$$\left. \frac{\partial L}{\partial \lambda_j} \right|_{(\mathbf{x}^*, \boldsymbol{\lambda}^*)} = b_j - h_j(\mathbf{x}^*) = 0, \quad j = 1, \dots, q$$

'and so

$$L(\mathbf{x}^*, \boldsymbol{\lambda}^*) = f(\mathbf{x}^*),$$

demonstrating that the point $\mathbf{x}^*$ is a local optimum of $f(\mathbf{x})$ if the point $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ is a local optimum of $L(\mathbf{x}, \boldsymbol{\lambda})$.

The method of Lagrange multipliers

Theorem 5.3 (First-order necessary conditions for local optimality) If the function $f(\mathbf{x})$ in (5.20) and the functions $h_1, \dots, h_q$ in (5.21) are all twice differentiable with respect to all pairs of their argument components, and if $f(\mathbf{x}^*)$ is a local optimum of the non-linear optimisation problem (5.20)–(5.22), then the conditions in (5.25) are satisfied at the point $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ for some vector $\boldsymbol{\lambda}^* \in \mathbb{R}^q$.

Theorem 5.4 (Second-order sufficient conditions for global optimality) If $f(\mathbf{x})$ is a concave function and if the functions $h_1, \dots, h_q$ are all convex functions, then any point $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ satisfying the conditions in (5.25) is a global maximum $\mathbf{x}^*$ of the non-linear optimisation problem (5.20)–(5.22).

Example 5.3 (pp. 171–172)

Consider the problem of

$$\text{minimising } z = 2x_1^2 + x_2^2 - x_1x_2 - 8x_1 - 3x_2$$

subject to the constraints

$$\begin{aligned} 3x_1 + x_2 &= 7, \\ (x_1, x_2) &\in \mathbb{R}^2. \end{aligned}$$

Example 5.3 (pp. 171–172)

In order to take advantage of the result of Theorem 5.4, we maximise the function $f_{4.2}(x_1, x_2) = -2x_1^2 - x_2^2 + x_1x_2 + 8x_1 + 3x_2$ instead, subject to the constraints (5.28)–(5.29). The gradient vector of $f_{4.2}$ is $\nabla f_{4.2}(\mathbf{x}) = [-4x_1 + x_2 + 8 \quad -2x_2 + x_1 + 3]^T$, from which it follows that its Hessian matrix is

$$\nabla^2 f_{4.1}(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f_{4.2}}{\partial x_1^2} & \frac{\partial^2 f_{4.2}}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f_{4.2}}{\partial x_2 \partial x_1} & \frac{\partial^2 f_{4.2}}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} -4 & 1 \\ 1 & -2 \end{bmatrix}. \quad (5.30)$$

$$L_{4.1}(x_1, x_2, \lambda) = -2x_1^2 - x_2^2 + x_1x_2 + 8x_1 + 3x_2 + \lambda(7 - 3x_1 - x_2).$$

Setting the partial derivatives of $L_{4.1}$ with respect to x_1 , x_2 and λ equal to zero yields the system of linear equations

$$\begin{aligned} \frac{\partial L_{4.1}}{\partial x_1} &= -4x_1 + x_2 + 8 - 3\lambda = 0, \\ \frac{\partial L_{4.1}}{\partial x_2} &= -2x_2 + x_1 + 3 - \lambda = 0, \quad \text{and} \\ \frac{\partial L_{4.1}}{\partial \lambda} &= 7 - 3x_1 - x_2 = 0. \end{aligned}$$

We solve this system by performing Gauss-Jordan elimination (see Algorithm 1.1) on the augmented matrix

$$\mathbf{X}^{(1)} = \left[\begin{array}{ccc|c} -4 & 1 & -3 & -8 \\ 1 & -2 & -1 & -3 \\ -3 & -1 & 0 & -7 \end{array} \right].$$